Generating Probabilistic Aerodrome Forecasts (TAFs) from Numerical Weather Prediction: A Benchmark-Driven Model Review

Spanish airport network, 2020–2025

Working review

Abstract

We study whether machine learning can produce skilful, well-calibrated aerodrome forecasts (TAFs) from numerical weather prediction (NWP) and recent observations, and beat the official human-issued TAF. Using a strictly causal feature frame over 24 Spanish airports (2.5 M METARs, 0.25 M TAFs; ERA5 reanalysis as a perfect-prognosis NWP proxy), we benchmark a ladder of tabular models and a family of sequence/probabilistic models (GRU, LSTM, Transformer, TCN, and a Temporal Fusion Transformer with quantile outputs) on a frozen 2025 test set. The best model is a logistic *stacked ensemble* of {MLP, gradient boosting, TFT} reaching **BSS** +0.348 / **HSS** 0.513, versus the official TAF's −0.108, and it is near-perfectly calibrated. Driving PROB/TEMPO group generation with it yields an operational TAF at BSS +0.215. Architecture is *not* the limiting factor: data scaling, model class, and the available feature set all plateau near BSS ≈ +0.33, indicating a data ceiling set by the ingested NWP fields and the perfect-prognosis setup.

1 Objective

Predict, from information available at issue time T_0, the hourly aerodrome state over the TAF validity window *probabilistically*, so the model can be scored fairly against the official TAF and drive automatic PROB/TEMPO change-group generation. The headline event is **IFR-or-worse** (sub-instrument ceiling/visibility) — operationally critical and rare (∼3.6%).

2 Data and setup

Targets are METAR-derived vis, ceiling, wind and flight category at valid time t. Features are strictly causal (enforced structurally): current and lagged METAR state (t_0, $-1/-3/-6$ h, with tendencies); ERA5 at t and T_0 and the $T_0 \to t$ tendency (perfect-prognosis NWP); lead time; cyclical hour/day; static airport attributes. Splits are temporal and blocked: train $<$ 2024, validation = 2024, frozen test = 2025; calibration is fit on validation only.

3 Metrics

BSS (Brier Skill Score of P(IFR-or-worse) vs climatology; hedging-aware, bootstrap CIs); **HSS** (categorical skill of the adverse decision at a validation-tuned threshold); **reliability** (calibration). References: climatology (BSS = 0) and the official TAF (skyline). A key lesson: calibration (isotonic on validation) is decoupled from the decision threshold; the raw class-weighted argmax must not drive the forecast category.

4 Models tested

Baselines (persistence, climatology, official TAF); a tabular ladder (linear, gradient boosting, a GPU multi-task MLP); sequence models (GRU, LSTM+attention, Transformer, TCN); a TFT with quantile heads; and ensembles (linear blend, logistic stack). All deep models share one framework, calibration contract and evaluation.

5 Results

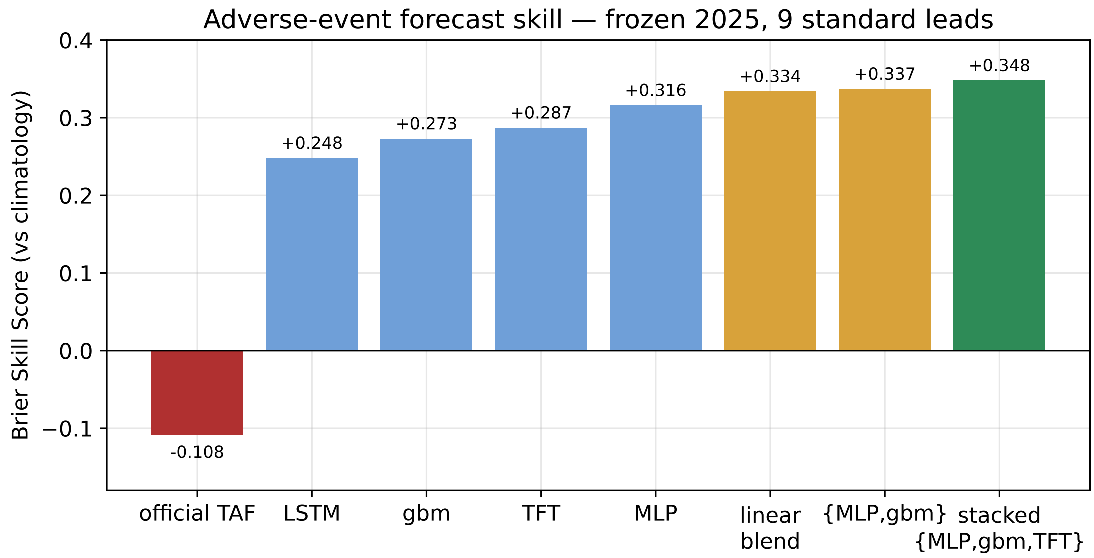

Figure 1: Adverse-event skill on the frozen 2025 test (9 standard leads). Every learned model beats the official TAF (−0.108); ensembling leads.

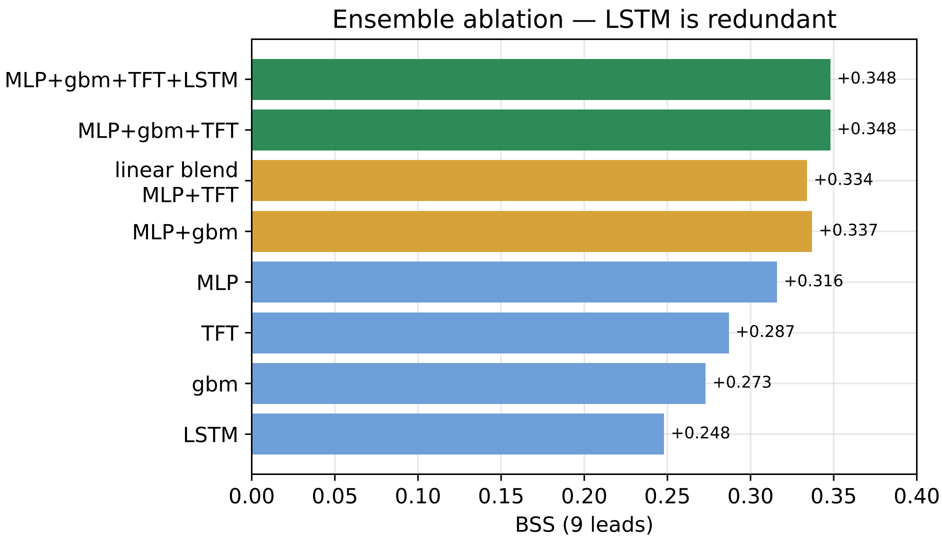

Figure 2: Stack ablation. {MLP, gbm, TFT} matches the full four-model stack (LSTM is redundant); the TFT adds +0.011 over the tabular-only {MLP, gbm} stack.

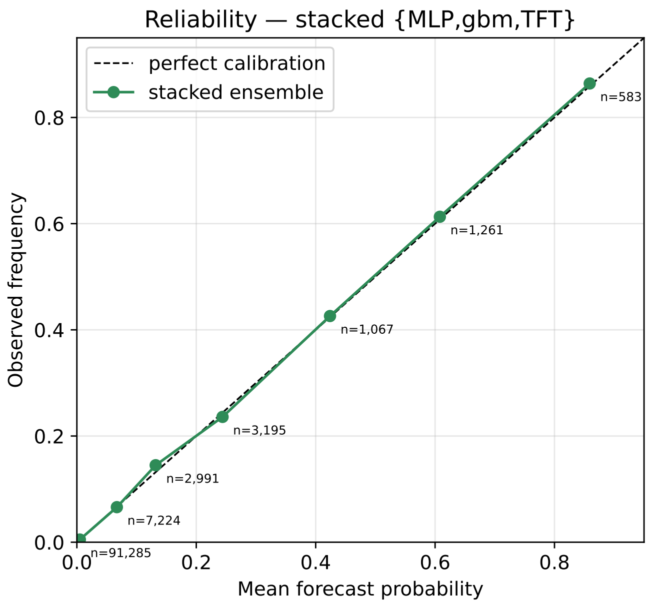

Figure 3: Reliability of the stacked ensemble: forecast probability matches observed frequency to within ±0.01.

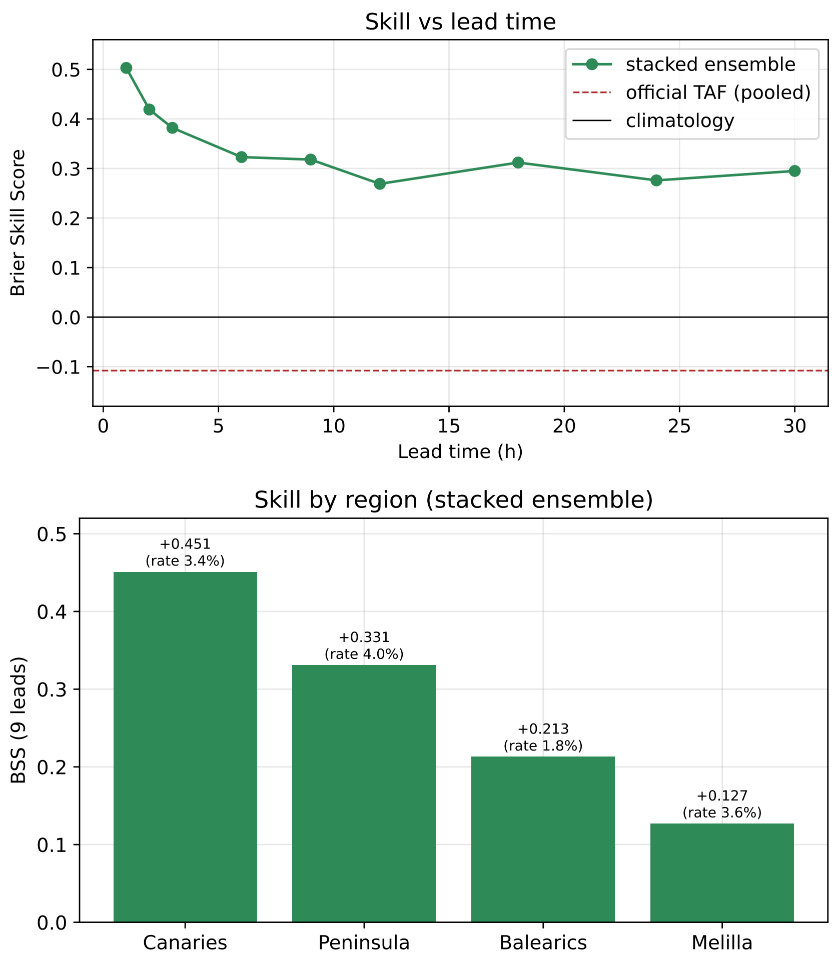

Figure 4: Skill vs lead time (top) and by region (bottom) for the stacked ensemble.

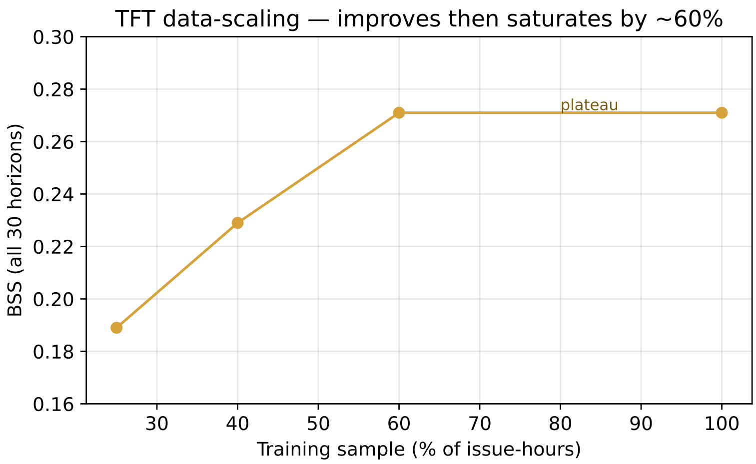

Figure 5: TFT data-scaling: skill improves then saturates by ∼60% sample.

6 Generating PROB/TEMPO TAFs

The stacked P(adverse) is quantized to the TAF buckets (none / PROB30 / PROB40 / firm) and combined with the TFT quantile element forecast to emit an hourly timeline with PROB/TEMPO groups. The regenerated operational TAF scores BSS +0.215 (vs −0.108). The TFT's visibility quantiles collapse to "clear" even at q_{10}, so adverse skill lives in the calibrated category probability, not the element distribution; adverse groups use category-representative conditions.

7 Discussion and limitations

Data volume, architecture and feature set all plateau near BSS ≈ +0.33 — a data ceiling, not a modelling one. ERA5's cloud-layer split (the obvious ceiling/fog predictor) was never ingested. The perfect-prognosis setup makes absolute skill optimistic; validation against a genuine IFS/ECMWF reforecast is the credibility milestone. The highest-value next step is richer / genuine-forecast NWP, not a larger network.

8 Conclusion

A calibrated, stacked ensemble of complementary model classes produces aerodrome forecasts substantially more skilful and better calibrated than the official TAF (BSS +0.348 vs −0.108) and drives a competitive operational PROB/TEMPO TAF.